

# Automation Exercise API Testing Project

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**38**

Test Cases

**33**

Passed

**5**

Failed

**86.8%**

Pass Rate

# + Agenda

01

## What is API Testing?

Definition, goals, and why it matters

02

## Tools & Technologies

Manual (Excel · Postman) vs Automation (Java · Rest Assured · TestNG · Maven)

03

## Why These Tools?

Comparison with alternatives — why not others

04

## Testing Strategy

Functional · Positive · Negative · HTTP Methods

05

## Test Case Design

38 TCs across 7 API modules — distribution

06

## Manual Testing — Postman

Request execution, validations, examples

07

## Automation — Rest Assured

Code structure, coverage, validations implemented

08

## Test Results & Defects

Allure Report · 5 failures · root cause analysis

## + What is API Testing?

- API testing verifies communication between software systems.
- Ensures APIs return correct responses for given requests.
- Validates functionality, reliability, performance, and error handling.

## + Project Goal

- To test the APIs of Automation Exercise website both manually (Postman) and automatically (Rest Assured + TestNG).

## How API Works



Client / Test Script



HTTP Request (GET / POST / PUT / DELETE)



API Layer — [automationexercise.com](https://automationexercise.com)



HTTP Response (status code + JSON body)



Assertion (200 OK? message correct?)

## + Tools & Technologies

### + Manual Testing

- Excel — Test Case Management (TC ID · Steps · Expected vs Actual · Pass/Fail)
- Postman — API Request Execution & Manual Validation

### + Automation Testing

- Java — Test Language (OOP, strongly typed, industry standard)
- Rest Assured — Java library for API automation
- TestNG — Test framework (grouping, parallel exec, data providers)
- Maven — Build & dependency management (single POM file)

### + Version Control

- Git - GitHub — Source control & collaboration



**Maven**



**TestNG**

## + Testing Strategy

### + Functional Testing

- Positive Testing — valid inputs, verify correct 200/201 response
- Negative Testing — invalid/missing params, verify error response

### + Validations

- Status Code Validation — 200, 201, 400, 401, 404, 405
- Response Body Validation — JSON field values
- Response Message Validation — 'User created!', 'invalid' etc.
- Input Parameter Validation — missing/empty/invalid fields

### Important Observation

- HTTP Status Code is always 200
- Actual result is determined by responseCode and message in the Response Body
- Therefore, validations were performed on both the HTTP response and the JSON body.

### + HTTP Methods Tested

#### GET

Retrieve product list, brands, user details

#### POST

Create account, login, search products

#### PUT

Update account information

#### DELETE

Delete user account



## + Test Case Design

| Module             | # TCs     |
|--------------------|-----------|
| Products API       | 4         |
| Brands API         | 2         |
| Login API          | 5         |
| Create Account API | 20        |
| User Details API   | 3         |
| Update Account API | 2         |
| Delete Account API | 2         |
| <b>TOTAL</b>       | <b>38</b> |

Create Account (18) Login (5) Products (4) User Details (3)  
Brands (2) Update Acct (2) Delete Acct (2)



**A total of 38 test cases covering positive & negative scenarios across 7 API modules.**

## + Manual Testing

### + Why Postman?

- Easy API request execution — no code needed
- Inspect full response (headers, body, status)
- Validate before writing automation
- Organize requests in collections

### + Activities Performed

- Request Execution (GET/POST/PUT/DELETE)
- Status Code Verification (200/201/400/404/405)
- Response Body & Message Validation
- Positive & Negative test scenarios

### Postman Collection Structure

|                |             |
|----------------|-------------|
| Products API   | 4 requests  |
| Brands API     | 2 requests  |
| Login API      | 5 requests  |
| Create Account | 20 requests |
| User Details   | 3 requests  |
| Update Account | 2 requests  |
| Delete Account | 2 requests  |

## + Project Structure

### ▼ Project-ITI

- > 📁 Products
- > 📁 Brands
- > 📁 Login
- > 📁 Create Account
- > 📁 User Details
- > 📁 Update Account
- > 📁 Delete Account

### ▼ 📁 Brands

- GET GET Brands List
- PUT PUT Brands List - Method Not Allowed

### ▼ 📁 User Details

- GET GET User Detail - Valid Email
- GET GET User Detail - Non-Existing Email
- GET GET User Detail - Missing Email

### ▼ 📁 Products

- GET GET Products List
- POST POST Search Product - Valid Keyword
- POST POST Products List - Method Not Allowed
- POST Post Search Product - Missing Parameter

### ▼ 📁 Login

- POST POST Verify Login - Valid Credentials
- POST POST Verify Login - Missing Email
- POST POST Verify Login - Invalid Email
- POST POST Verify Login - Invalid Password
- DEL Delete Verify Login - Method Not Allowed

### ▼ 📁 Update Account

- PUT PUT Update Account - Valid Data
- PUT PUT Update Account - Missing Password

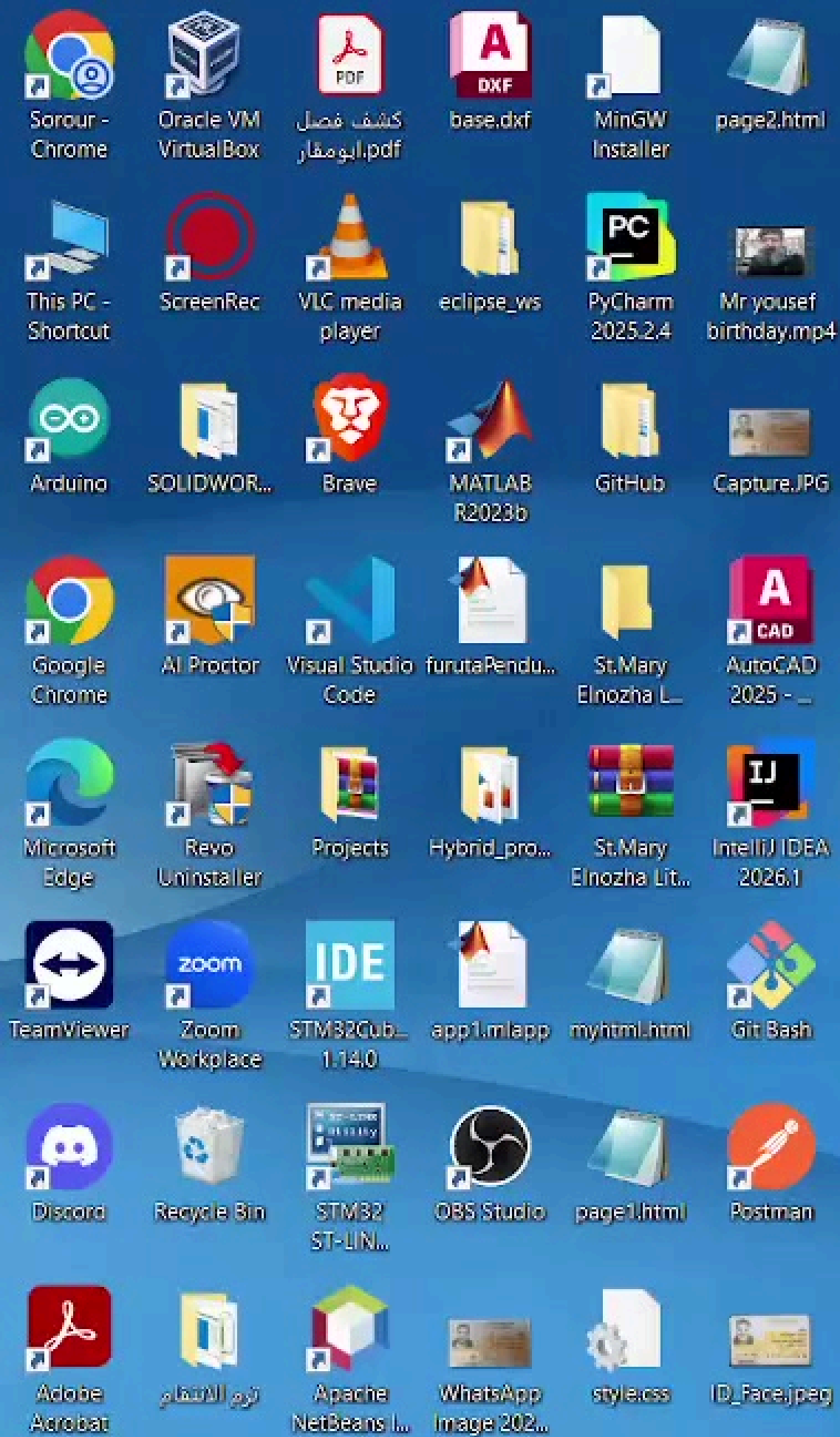
### ▼ 📁 Delete Account

- DEL DELETE Account - Valid Credentials
- DEL DELETE Account - Invalid Password

### ▼ 📁 Create Account

- POST POST Create Account - Valid Data
- POST POST Create Account - Missing Password
- POST POST Create Account - Missing FirstName
- POST POST Create Account - Missing LastName
- POST POST Create Account - Missing Mobile
- POST POST Create Account - Missing City
- POST POST Create Account - Missing State
- POST POST Create Account - Missing Zipcode
- POST POST Create Account - Missing Address1
- POST POST Create Account - Missing DOB
- POST POST Create Account - Missing Address2
- POST POST Create Account - Missing Company
- POST POST Create Account - Missing Title
- POST POST Create Account - Arabic FirstName
- POST POST Create Account - Arabic LastName
- POST POST Create Account - Special Chars FirstName
- POST POST Create Account - Special Chars LastName
- POST POST Create Account - Alphabetic Mobile





Activate Windows  
Go to Settings to activate Windows.

## + Automation Testing

### + What is Rest Assured?

Java library for REST API automation — provides a fluent BDD-style DSL: `given()` / `when()` / `then()`

### + Automation Coverage

- 38 automated test cases
- 7 API modules
- Positive & negative scenarios
- `baseUrl` configured once in `BaseTest` class

### + Validations Implemented

- Status Code · Response Body · Message

## Project Code Structure



`src/test/java`

Root test package



`BrandsApiTests.java`

GET brands list · PUT method not allowed



`CreateAccountApiTests.java`

18 TCs: valid, missing fields, invalid input



`DeleteAccountApiTests.java`

DELETE valid creds · invalid password



`LoginApiTests.java`

5 TCs: valid, missing email, invalid creds



`ProductsApiTests.java`

GET list · POST search · method not allowed



`UserDetailsApiTests.java`

GET by email: valid, non-existent, missing



`UpdateAccountApiTests`

Put different data : valid, non-existent, missing



`BaseTest.java`

Base URL + shared setup (`@BeforeSuite`)



`pom.xml`

Maven: Rest Assured · TestNG · Allure deps

## + Why These Tools — Not the Alternatives

### Rest Assured vs HTTPClient / Retrofit

#### ✓ We used: Rest Assured

- Fluent BDD-style DSL (given/when/then)
- Built-in JSON path assertions
- Native TestNG integration
- Less boilerplate than raw HTTP clients

#### ✗ Not: HTTPClient / Retrofit

- Manual JSON parsing required
- No built-in assertion DSL
- More boilerplate code

### TestNG vs JUnit

#### ✓ We used: TestNG

- Built-in data providers for DDT
- Test grouping & dependencies
- Parallel execution out of box
- Richer annotations (@BeforeSuite)

#### ✗ Not: JUnit

- No built-in parallel execution
- Weaker test grouping
- Needs extension for DDT

### Maven vs Gradle

#### ✓ We used: Maven

- Industry-standard XML POM file
- Simpler dependency management
- Wide IDE support (IntelliJ/Eclipse)
- Stable, well-documented lifecycle

#### ✗ Not: Gradle

- Groovy/Kotlin DSL learning curve
- Overkill for simple projects
- More suitable for large and complex projects

# + Test Results & Defects

38

Total TCs

38

Executed

33

Passed

5

Failed

86.8%

Pass Rate

| API Module         | Pass | Fail |
|--------------------|------|------|
| Brands API         | 2    | —    |
| Create Account API | 15   | 5    |
| Delete Account API | 2    | —    |
| Login API          | 5    | —    |
| Products API       | 4    | —    |
| Update Account API | 2    | —    |
| User Details API   | 3    | —    |
| TOTAL              | 33   | 5    |

## Failed Test Cases — Create Account API

- Special Characters in First Name — API accepted @@@@ (expected 400)
- Special Characters in Last Name — API accepted ##### (expected 400)
- Alphabetic Mobile Number — API accepted 'abcdf' (expected 400)
- Special Chars in Mobile — API accepted #\$\$@ chars (expected 400)
- Spaces in Mobile Number — API accepted '012 701' (expected 400)

Root Cause: Missing input sanitization on name & mobile fields

# + Allure Report Overview



Allure Report

June 25, 2026 at 3:25:26 AM (20s 435ms)

86.84%

Results 38

Categories 1

Quality Gates 0

Global Attachments 0

Global Errors 0

Total

38

Failed 5

Passed 33

Search tests

Retry

Flaky

Transition

Categories

Total 38

Failed 5

Passed 33

Sort by: Order Earliest

Surefire suite

533

Surefire test

533

BrandsApiTests

2

CreateAccountApiTests

515

1 Validate POST Create Account with alphabetic mobile number returns error

499ms

2 Validate POST Create Account with spaces in mobile number returns error

399ms

3 System should reject special characters in mobile number field

848ms

4 Validate POST Create Account with special characters-only firstname returns error

385ms

|                                                                                       |       |
|---------------------------------------------------------------------------------------|-------|
| ▼ Surefire suite                                                                      | 5     |
| ▼ Surefire test                                                                       | 5     |
| ▼ CreateAccountApiTests                                                               | 5     |
| ✖ 1 Validate POST Create Account with alphabetic mobile number returns error          | 499ms |
| ✖ 2 Validate POST Create Account with spaces in mobile number returns error           | 399ms |
| ✖ 3 System should reject special characters in mobile number field                    | 848ms |
| ✖ 4 Validate POST Create Account with special characters-only firstname returns error | 385ms |
| ✖ 5 Validate POST Create Account with special characters-only lastname returns error  | 369ms |

▼ allure-results

0c619e02-87c0-48c7-af11-6a6a415b2cb5-container.json

2a6b1592-753b-4e23-9685-e8ab09d24200-result.json

2b1e7e11-b053-46b9-bf56-002505b77a2e-result.json

2b4cbd6a-1041-4d83-b9dc-3a28d97ce817-result.json

2fb60bb6-6b54-4333-9d51-c4ad858b7e8f-result.json

3cf2ff02-3bc5-41cf-ba34-549a43452ef1-result.json

04bccb59-23c8-45ca-9263-e5d331c4e841-result.json

5b1d3915-6a24-4ef2-90e0-5e300faac7a3-result.json

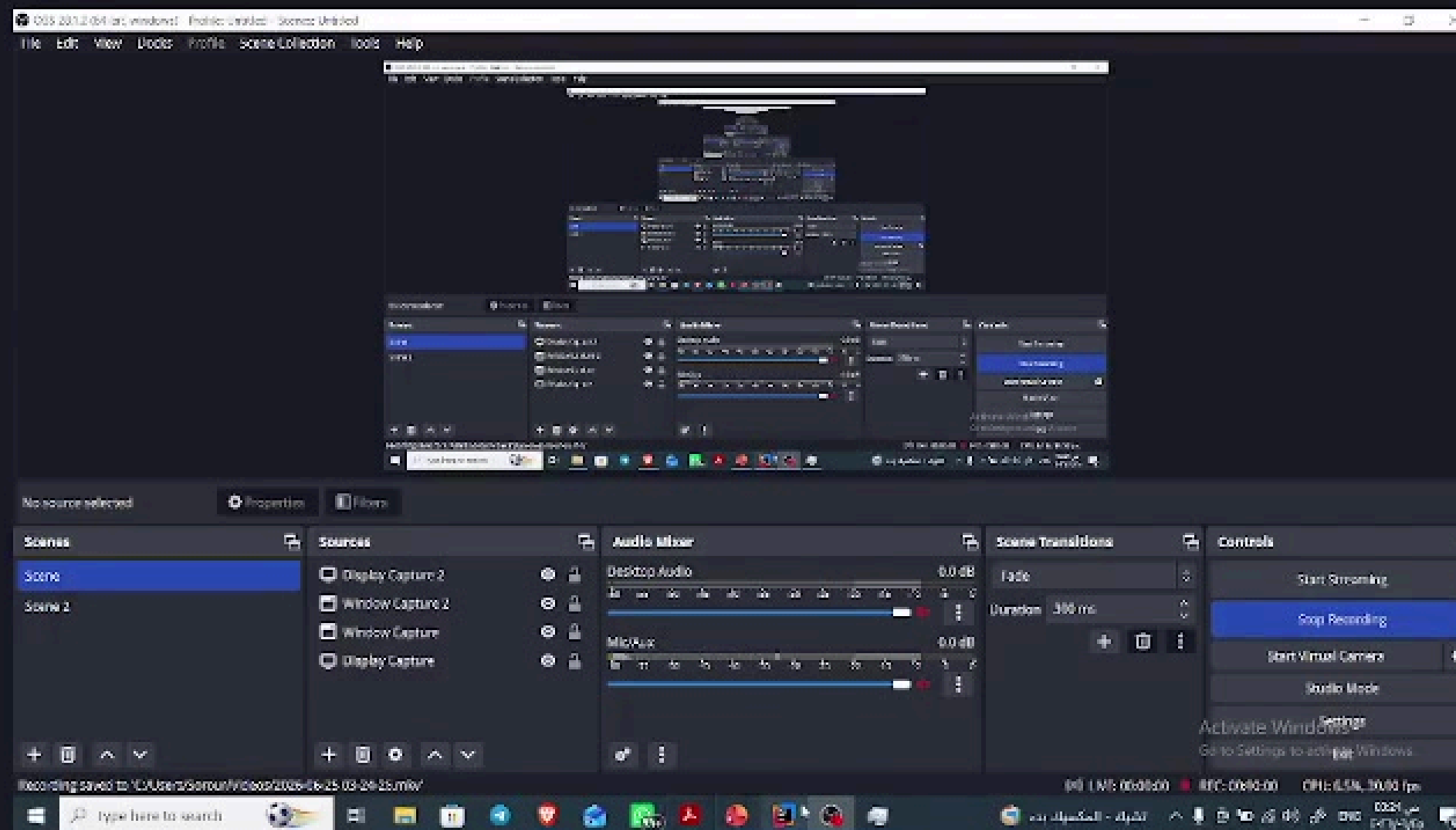
5da9d9ef-34c3-4907-bda9-f0e82eb33178-result.json

5fe8a00a-532c-4da1-a846-cfd120ee9071-result.json

12e098fa-85b6-4e72-8a94-8c2820a1cbe7-result.json

37ebfd59-cab3-4f18-803e-596e25f69f8d-result.json





No source selected

Properties

Filters

Scenes

Scene

Scene 2



Sources

Display Capture 2

Window Capture 2

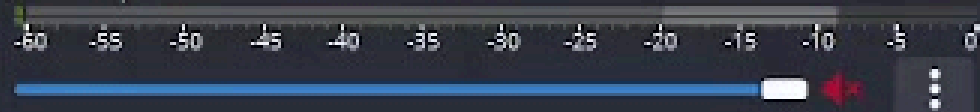
Window Capture

Display Capture

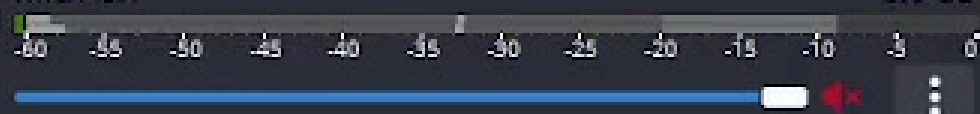


Audio Mixer

Desktop Audio



Mic/Aux



Scene Transitions

Fade

Duration 300 ms



Controls

Start Streaming

Stop Recording

Start Virtual Camera

Studio Mode

Activate Windows

Go to Settings to activate Windows.

Recording saved to 'C:/Users/Sorour/Videos/2026-06-25 03-24-26.mkv'

LIVE: 00:00:00 REC: 00:00:01 CPU: 6.5%, 30.00 fps



Type here to search



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03:24 ص  
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+ GitHub Repo



- TestCases report

# Thank you so much

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